

Working with atlas-transferred colour–texture data: from colour morphospaces to an expert-in-the-loop pattern tool

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Abstract

We have a tool that bakes per-specimen colour textures onto a shared mean/atlas mesh, turning a set of 3D-scanned organisms into an aligned, population-scale colour–texture dataset. This report addresses the follow-on question: *how does a biologist actually work with that data?* We trace a deliberate progression. A colour morphospace (PCA on colour composition) solves the easy, high-variance factors — hue clusters separate cleanly and auto-cluster correctly — but goes no further. Subtle *pattern* (stripe count, texture) is invisible to colour, so we introduce a general, CPU-only graph-spectral / GFT descriptor that raises the recoverable signal substantially. Yet a deeper limit remains: *purely unsupervised* discovery cannot surface low-variance, overlapping, or rare structure — this is a property of high-dimension–low-sample-size geometry, not a tuning failure, and a trustworthy pipeline must therefore *abstain* rather than fabricate. The productive resolution is a small amount of expert supervision. We show that multi-label *tagging* — the expert names traits and clicks specimens that have them — is far more label-efficient than pairwise similarity feedback and yields a directly interpretable classifier latent. We describe the expert-tagging tool that resulted, justifying each design choice with the underlying experiments, and summarise the full study in an appendix.

1 Introduction

The upstream tool performs *atlas texture transfer*: every specimen’s surface colour is resampled onto a single shared mean mesh with a common UV layout, so that a fixed face (or region) refers to the same anatomical location across all individuals. The output is a registered (specimens $\times$ faces $\times$ colour) tensor — a population in correspondence. This is genuinely new data for a lab, and it invites an obvious question that turns out to be the hard part: *what analysis lets a non-technical biologist see the population’s structure, with minimal UI?*

This report follows the path the project actually took:

1. **Colour only goes so far.** A colour morphospace already solves the high-variance hue factors (Sec. 2); it does not touch pattern.
2. **Pattern needs structure-aware descriptors.** Graph-spectral / GFT features on the mesh recover pattern that colour PCA cannot (Sec. 3).
3. **Unsupervised discovery has a hard ceiling.** Even with the right descriptor, subtle/overlapping/rare structure is supervised-recoverable but *not* unsupervised-discoverable (Sec. 4).
4. **Expert supervision works.** Multi-label tagging breaks the ceiling cheaply and is what we shipped (Secs. 5–6).

Datasets. We develop and validate on two populations. *Fishy* is a synthetic set of $n=250$ textured fish with four *known* planted factors, used as ground truth: *belly* hue (2 groups), *tail* hue (2 groups), *stripe* count (4 vs. 5, an 80/20 split), and rare *rosy cheeks* (5.6%, present on only 14/250). *Mussels* is a real population of $n=31$ scanned shell valves with no ground truth, used to check that conclusions transfer to real, small- n data. Everything is CPU-only and assumes the unlabelled regime; ground-truth labels are used for validation only, never to colour a user-facing plot.

2 Colour is easy — and already solved

The existing colour morphospace (PCA on an area-weighted colour histogram, or per-region mean colour) already recovers the high-variance colour factors essentially perfectly. On fishy, belly and tail hue land on PC1/PC2 as four clean quadrants; balanced cross-validated classification accuracy is ≈ 1.00 for both, and an automatic GMM+BIC clustering of the colour plane recovers the four belly $\times$ tail groups at **ARI** = 0.99 with no tuning (Fig. 1). On the real mussels the same machinery finds an interpretable periostracum-darkness split. *This part needs no change.* The lesson that matters is the flip side: colour PCA scores stripe at only **0.64** and cheeks at **0.48** ($\approx$ chance). The high-variance axes a colour morphospace surfaces are exactly the *nuisance* ones (base hue, belly/tail colour); the biologically interesting pattern variation contributes little total variance and is buried.

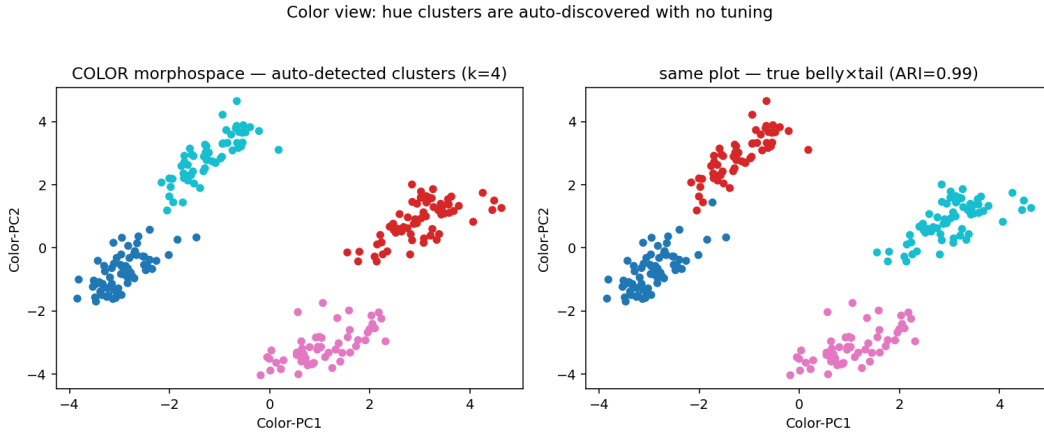


Figure 1: The colour morphospace already solves the easy factors: belly $\times$ tail hue form four clean groups that an automatic clusterer recovers at ARI 0.99. Subtle pattern (stripe count) and the rare cheeks variant are *not* here — they carry little variance and stay hidden.

3 Pattern needs a structure-aware descriptor

Stripe count is a *geometric*, shift-jittered feature: a colour-composition histogram is blind to it, and even spatial colour vectors are smeared because stripe positions vary specimen-to-specimen. The fix is a descriptor that encodes *layout*, not just composition. We settled on a **graph Fourier transform (GFT)** representation: treat each colour channel as a signal on the mesh graph and project it onto the low eigenmodes of the mesh Laplacian (“sines and cosines on the surface”). Signed GFT coefficients are general (no pattern-specific assumptions), mesh-intrinsic (UV/topology-independent, which matters for real meshes), and CPU-trivial. A shared-codebook *texton* histogram — a texture vocabulary learned from the millions of pooled local patches across all specimens — pushes stripe a little further still.

Table 1: Balanced cross-validated accuracy per factor (chance ≈ 0.5), by descriptor. Colour PCA solves the hue factors but not pattern; the GFT/texton descriptors recover stripe and lift cheeks, with *no* hand-coded body axis. Pattern is recoverable by a *classifier* — it still does not appear on the leading unsupervised axes (Sec. 4).

factor	colour PCA	GFT coeff. ($k=40$)	texton BoW	<i>best supervised</i>
belly	1.00	1.00	1.00	1.00
tail	1.00	1.00	1.00	1.00
stripe	0.64	0.93	0.97	0.97
cheeks	0.48	0.77	0.54	~ 0.79

A consistent, important subtlety surfaced here and recurs throughout: **the discriminative signal lives in the low-variance tail of the representation**. Figure 2 sweeps the number of PCA dimensions. Colour classification saturates by ≈ 4 dimensions; stripe accuracy keeps climbing to ≈ 30 dimensions, while a 95%-explained-variance criterion would stop at ≈ 12 . *Explained variance under-counts the dimensions the subtle factor needs*. This single observation justifies a core demo choice — keep ≈ 24 PCA dimensions, sized by the hardest factor, not by a variance threshold (which would silently discard stripe). The same plot shows the wall ahead: stripe’s unsupervised cluster ARI is ≈ 0 at *every* dimensionality.

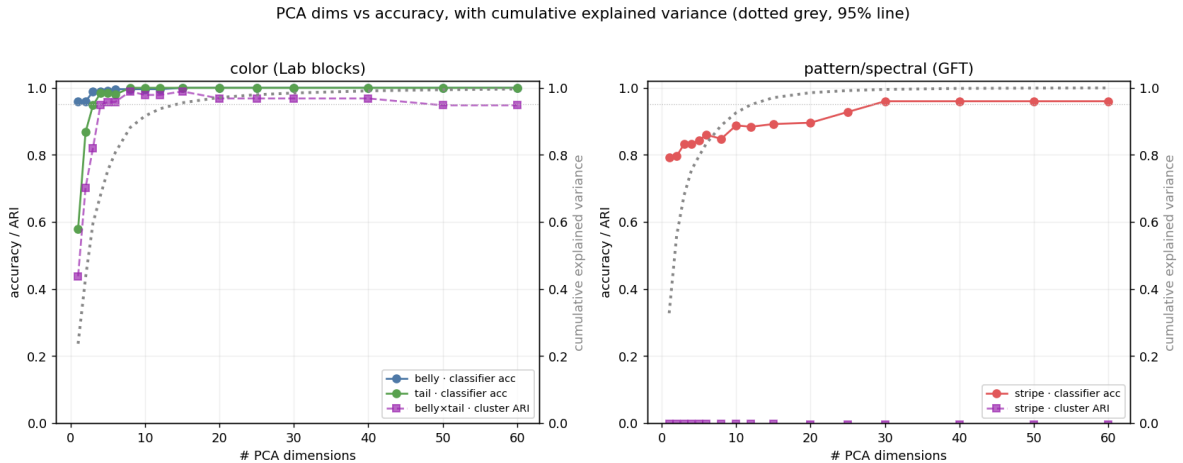


Figure 2: Accuracy vs. PCA dimensions, with cumulative explained variance (dotted). *Left:* colour classification saturates by ~ 4 dims; the belly $\times$ tail clustering ARI is high. *Right:* stripe accuracy needs ~ 30 dims — well past the 95%-variance line (~ 12) — so a variance criterion drops it; and stripe never *clusters* (purple ≈ 0 everywhere). Motivates keeping ~ 24 dims and treating stripe as a supervised, not unsupervised, target.

4 The unsupervised wall: discovery is not measurement

A better descriptor raises the *supervised* ceiling but does not make subtle structure *discoverable*. Across PCA, ICA, UMAP, projection pursuit, and density clustering, the result is the same: a variance-driven morphospace surfaces only factors with large between-specimen variance. Stripe (4-vs-5 under position jitter, 80/20) has essentially *no density gap* — the optimal projection’s dip statistic sits at the noise floor — so it is recoverable by a classifier (≈ 0.9) yet invisible to clustering (ARI ≈ 0). The rare cheeks class ($n=14$) is below the count that forms a significant mode at all. This is not a defect of our pipeline; it is the expected behaviour of high-dimension-low-sample-

size (HDLSS) “spiked covariance” geometry: when a factor’s variance is below the per-face noise variance, the leading eigenvector rotates away from it. *Supervised-recoverable but not unsupervised-discoverable* is predictable in advance.

The honest engineering response is a **trustworthy gate** that finds real structure and refuses to invent it. Combining a SigClust Gaussian-null test, a covariance-preserving consensus null, and per-cluster bootstrap stability, the pipeline confirms the colour clusters (ARI 0.99 at $n=250$; 0.83 at $n=40$) and *correctly abstains* on the pattern descriptor at $n=40$ rather than reporting spurious groups. On the real mussels the gate likewise reports a continuous ring-contrast gradient, not fake clusters. A morphospace you can publish from is one that knows what it cannot see — which is precisely why, for the subtle factors, we turn to the expert.

5 Expert tag-supervision breaks the ceiling

If the structure is real but not density-discoverable, a small amount of supervision is the lever. We evaluated several human-in-the-loop channels (pairwise must-link/cannot-link constraints, anchor-and-rank triplet feedback, spring-energy morphospace morphing; see Appendix E) and found two robust facts. First, a genuinely overlapping low-variance factor **cannot be separated in a 2-D layout at all** — even an oracle ranking by true stripe count caps at ~ 0.56 neighbourhood purity in 2-D, because 2-D has no room for it. Second, **multi-label tagging is markedly more label-efficient than pairwise feedback**: the expert creates named tags (“green tail”, “5 stripes”) and clicks the specimens that have them, then a per-tag classifier *places* every unlabelled specimen and the population is laid out by the classifier’s own latent.

Figure 3 compares classifier families for this “placement” job. A **simple per-tag linear logistic** model is the best placer — it reaches 8-way ARI 0.86 at 80 tagged specimens (about the same recovery that needed ~ 200 – 400 *pairs* via constraint feedback) and per-factor accuracy 0.93 at just 40 tags. Prototype/NCM is a close, tiny-budget-robust second; semi-supervised label propagation and learned 2-D embeddings *under-perform*, because the raw-descriptor neighbourhood graph is dominated by colour, so labels propagate along the wrong manifold. The factors are close to linearly separable in the 24-PC space, so the linear model both places best and keeps the demo fast and interpretable.

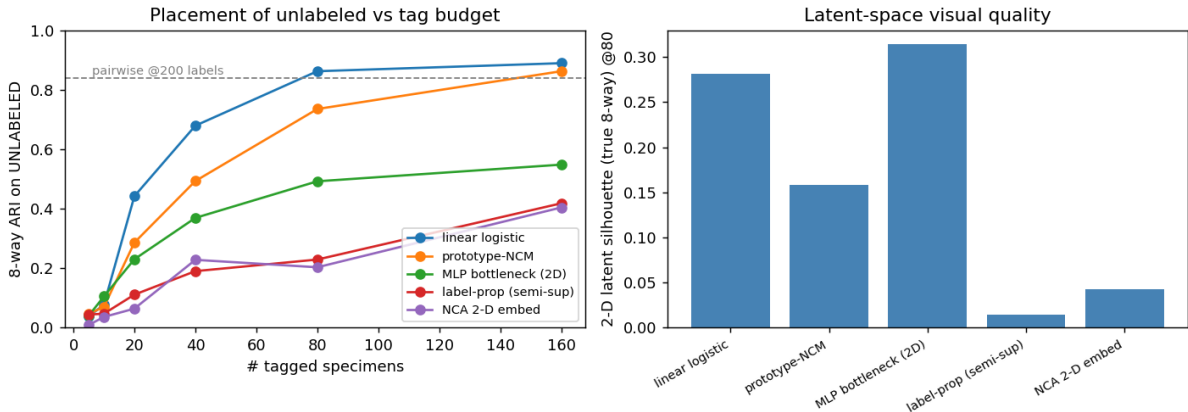


Figure 3: Choosing the classifier behind tag mode. *Left:* placement quality (8-way ARI on the *unlabelled*) vs. tag budget — a per-tag linear logistic model leads and matches, at 80 tags, the recovery that pairwise feedback reached only at ~ 200 labels (dashed). *Right:* 2-D latent visual quality; the linear model’s latent is nearly as clean as a purpose-built 2-D bottleneck while placing far better.

The classifier defines the latent; we still have to *display* it. The tag-score latent projected with **UMAP** roughly doubles the visual separation of a PCA projection (8-way silhouette 0.57 vs. 0.37 at 80 tags; Fig. 4). Crucially, UMAP on the *raw* descriptor with no tags shows no structure at all (silhouette -0.07): the tags-then-classifier step is what *creates* the structure; UMAP only lays it out. Because UMAP is stochastic and we want the map to react after every batch without teleporting, we re-fit it each batch *warm-started* from the previous embedding (init = previous layout, ~ 60 epochs). This is $\sim 2.5\times$ steadier than a naive re-fit, reacts from the very first batch, and needs no GPU or extra dependency — a parametric-UMAP encoder is marginally smoother but pulls in TensorFlow, and offline aligned-UMAP is far too slow for live use.

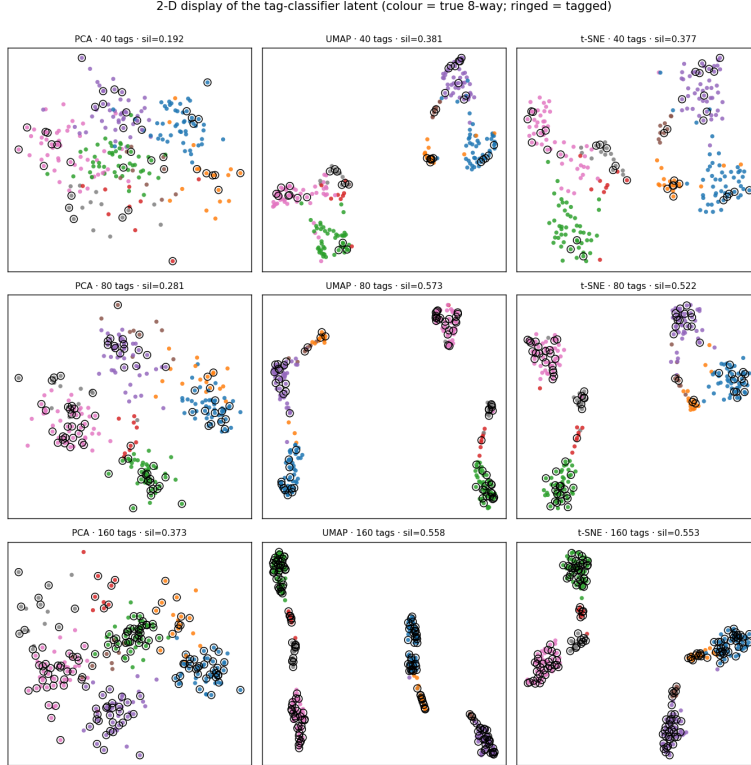


Figure 4: Displaying the tag-classifier latent in 2-D (rows: 40/80/160 tags; columns: PCA, UMAP, t-SNE; colour = true 8-way group). UMAP (and t-SNE) resolve the groups into clean islands where PCA stays a muddy blob. The structure comes from the tags; the projection only arranges it.

Active learning and the rare class. Two more choices follow from the data. After each applied batch the tool *auto-rolls* the next batch by **uncertainty** (the unlabelled specimens whose per-tag predictions are nearest 0.5); this beats random selection decisively (8-way ARI 0.96 vs. 0.81 at 80 labels; Fig. 5, left). But uncertainty sampling is a poor way to *first* encounter a rare class. For cold start we seed the very first batch by **novelty** — mean k -nearest-neighbour distance in the compressed PCA space — which surfaces the first rare “cheeks” specimen at rank 1 versus ~ 16 random labels, a $10\text{--}16\times$ speedup (Fig. 5, right). Novelty/anomaly scoring surfaces any *separable* minority (it does nothing for an arbitrary random rare subset), and only in the compressed space — distance-based anomaly collapses in the raw 640-D colour descriptor. Finally, the per-tag logistic uses `class_weight=’balanced’`, which adds ≈ 0.09 balanced accuracy for the rare factors in the early regime and ties SMOTE/ADASYN without the dependency.

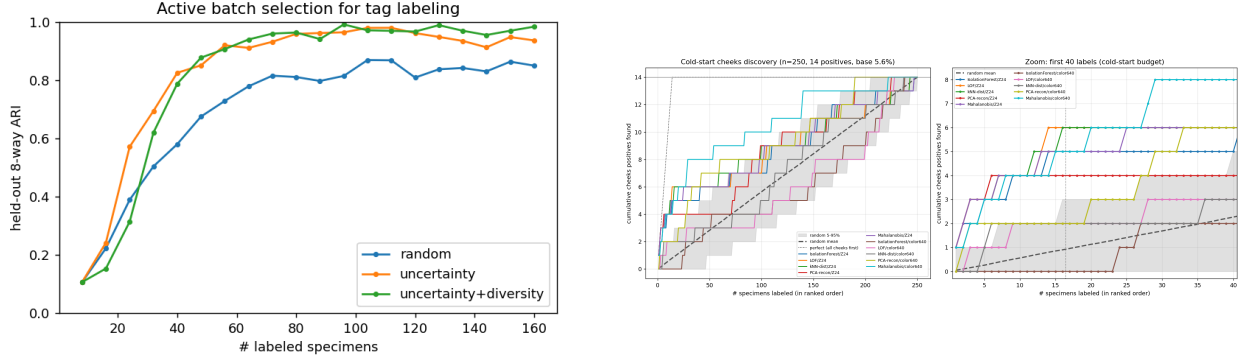


Figure 5: *Left:* after each batch, auto-rolling the next batch by classifier uncertainty $\gg$ random selection. *Right:* cold-start novelty seeding finds rare-class members an order of magnitude faster than random for the first few labels (then hands off to uncertainty sampling).

6 The expert-tagging tool

These findings assemble into a single interactive tool (Fig. 6). The expert explores batches of specimens, creates trait tags, selects an active tag and clicks the specimens that show it (multi-label), and hits *Apply*; the tool retraines the per-tag classifier and updates a 2-D morphospace and a clustering tree of the classifier latent. It then auto-rolls the next, most-informative batch. A frontend-only static build runs entirely in the browser for sharing.

Table 2: What landed in the tool, and the experiment that justifies it.

Choice	Why	Rejected alternative
Per-region colour+texture $\rightarrow$ PCA-24	subtle factors live in the low-variance tail; ~ 24 –30 dims keeps them	EV-threshold (~ 12 dims, drops stripe)
Per-tag linear logistic, $C=0.5$	best placer; near-linearly separable; fast & interpretable	MLP / label-prop / NCA (worse placement)
<code>class_weight='balanced'</code>	free +0.09 for rare tags early	SMOTE/ADASYN (tie, extra dep.)
Warm-started UMAP of tag latent	doubles separation over PCA; smooth, reactive, no GPU	PCA (muddy); parametric (TF); aligned (slow)
Uncertainty auto-roll	$\gg$ random (0.96 vs 0.81)	random / uncertainty+diversity (worse early)
Cold-start novelty seed	10–16 $\times$ faster first rare-class hit	NN-expansion / raw-space anomaly (fail)
Ward tree on the latent	label-free overview of recovered groups	—
Multi-label tagging	far more label-efficient than pairwise; direct predictions	pairwise / triplet / spring (2-D ceiling)

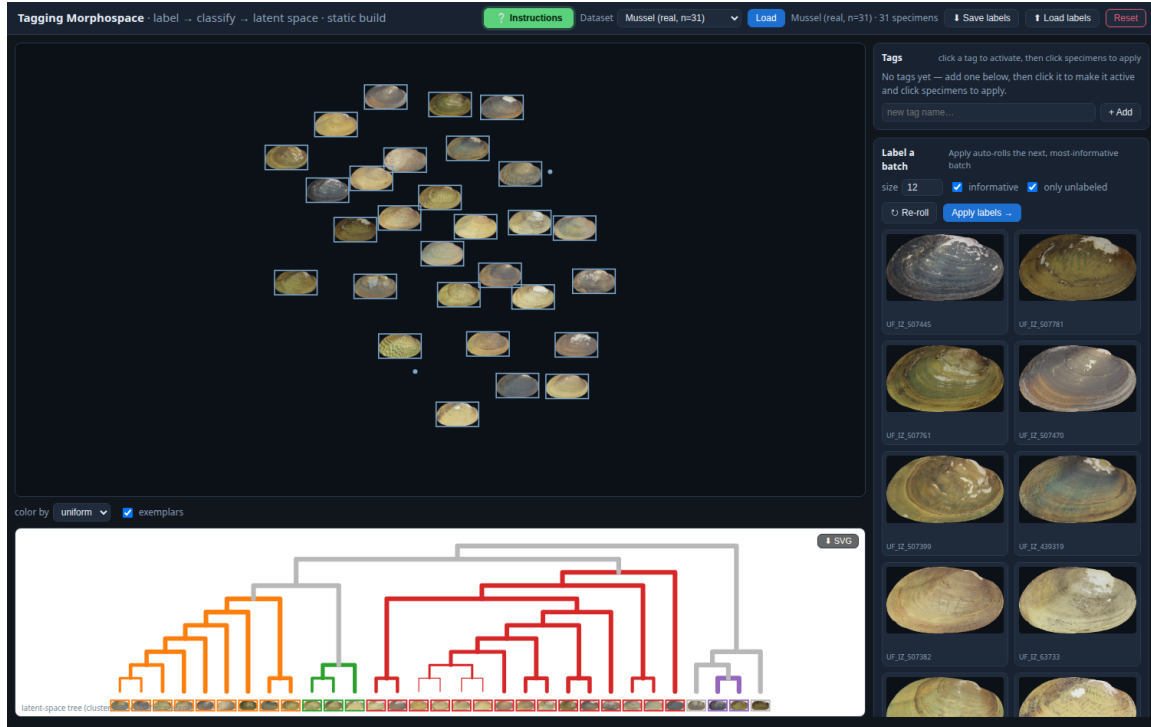


Figure 6: The expert-tagging tool (here on the real mussel valves). Left: the classifier-latent morphospace with specimen exemplars and a clustering tree; right: the tag palette and the batch the expert is labelling. Tag, Apply, and the next informative batch rolls automatically.

7 Discussion and limitations

The arc is the message: a colour morphospace solves what is easy; a general spectral/texture descriptor makes pattern *recoverable*; but recoverable is not the same as *discoverable*, and on small real populations the honest unsupervised answer is often “a colour split and a continuous gradient, nothing more.” A few minutes of expert tagging converts the supervised-only structure into a navigable, predicted morphospace at a fraction of the labelling effort of pairwise feedback. Honest limits remain: the rare cheeks class plateaus near ≈ 0.66 balanced accuracy in the tool’s 24-dimensional representation even given all positives (≈ 0.79 with the richest descriptor; 14 examples is simply few), and a truly overlapping factor shows as a gradient, not a cluster, in any 2-D view — so the tool’s value is that you can *see and label* such structure, not that it auto-separates. The methods are CPU-only and transfer to real data (mussels) and to different meshes (the spectral descriptor is topology-independent). A condensed ledger of all experiments follows.

Appendix: condensed experiment ledger

A Problem framing & baseline (EXP-00,01,04,05)

Verified the planted ground truth (belly/tail independent; stripe 79/21; cheeks 5.6%). Reproduced the module’s colour morphospace: belly/tail ≈ 1.0 , stripe 0.5–0.75, cheeks $\approx$ chance. Established the governing principle — *discovery* $\neq$ *measurement*: variance-driven reduction shows only high-variance factors; subtle structure must be measured/supervised. End-to-end “recommended” pipeline: colour clusters (ARI 0.99) + a measured stripe axis (0.64→0.89) + a rare-variant flag.

B Pattern descriptors (EXP-02,03,07,20,22,25,27,30,53,54)

Hand-tuned axial band-count detector (0.88, but taxon-specific) $\rightarrow$ *general* graph-spectral band energy $\rightarrow$ **signed GFT coefficients** ($k=40$; stripe 0.93, cheeks 0.77; the best single general descriptor) $\rightarrow$ **shared-codebook texton BoW** (stripe 0.97, the best on fishy). Segmentation+spatial/Endler ties GFT and adds interpretable arrangement stats. Both descriptors measure *count*, not the spacing/width confound (supervised direction correlates $|r| \gtrsim 0.95$ with count — GFT 0.99, segmentation 0.95 — and $\lesssim 0.03$ with spacing). *Negative result*: segment-then-GFT is worse than either alone — do not chain them. Dimension/mode sweeps: keep ~ 24 PCA dims / $k \gtrsim 40$ modes; explained variance mis-sizes both.

C The unsupervised wall & trustworthy gating (EXP-11,12,13–15,32,38–41)

Mapped what unsupervised discovery can honestly reach: well-separated balanced clusters (belly/tail) YES; subtle overlapping structure (stripe) NO (no density gap; dip $\approx$ noise floor); rare minority (cheeks, $n=14$) NO; genuinely continuous variation $\rightarrow$ the gate ABSTAINS. The HDLSS reframing (Jung–Marron spiked covariance) makes this predictable in advance. A SigClust + consensus-null + bootstrap-Jaccard gate finds the colour structure (ARI 0.99/0.83 at $n=250/40$) and abstains on pattern at $n=40$. Localized, per-region gated clustering and covariation-based mesh parcellation auto-isolate the colour modules and emit a label-free “structure map,” but still cannot cluster the subtle/rare factors.

D Representation, dimensionality, imbalance (EXP-53,54,55,56,57,59)

Subtle factors sit in the low-variance PCA/spectral tail, so keep generous dimensionality. ICA $\equiv$ PCA for linear classifiers (not a recovery lever), though ICA cleanly isolates the strong colour factors for interpretability. For class imbalance, `class_weight='balanced'` $\approx$ SMOTE $\approx$ ADASYN — the binding constraint is the minority *count*, not the loss machinery — so use the free reweighting.

E Expert-in-the-loop supervision (EXP-29,33,34,35,36,37,42–45)

Plain few-label supervision and label propagation are label-inefficient on a generic high-dim descriptor. Pairwise constraint metric-warping recovers stripe (ARI 0 $\rightarrow$ 0.42 at 80 pairs); anchor-and-rank triplet feedback sharpens whatever trait the expert ranks (“you get what you rank”); a spring-energy morphospace gives a faithful interactive morph. Two robust lessons: a low-variance overlapping factor has a hard *2-D ceiling* (~ 0.56 even with an oracle ranking), and diverse panels aid exploration while *local* panels aid fine discrimination. Recovering the full $2^3=8$ -cluster structure needs graded target distances + a low-rank metric + clustering in the 24-D space (ARI ≈ 0.84 at 200 labels), shown as a factor grid rather than hoping 2-D springs separate eight groups.

F The tagging tool (EXP-46–52,58,60)

Multi-label tagging beats pairwise feedback for label efficiency and yields direct predictions. Best placer = per-tag linear logistic in 24-PC space (8-way ARI 0.86 at 80 tags); semi-supervised/embedding models underperform. Display = **warm-started UMAP** of the tag-score latent (doubles PCA’s separation; reactive from batch 1; no TF). Active **uncertainty** auto-roll $\gg$ random; **cold-start novelty** seeding finds rare-class members 10–16 $\times$ faster; anomaly scoring surfaces any *separable* minority, in the compressed space only. Net build: per batch = retrain six linear tag classifiers (ms) + a warm-started UMAP fit (~ 0.2 s at $n=250$) $\rightarrow$ client tweens to the new coordinates.

G Generality / real data (EXP-08,09,10,21,28,40,41)

The same engine transfers to the real mussel valves (different mesh, baking artefacts) and to a charisma-style phenetic dendrogram with mesh-rendered exemplars: it produces an interpretable colour split and a continuous ring-contrast pattern axis, and the gate correctly abstains from inventing discrete pattern clusters — the behaviour we want when ground truth is unavailable.